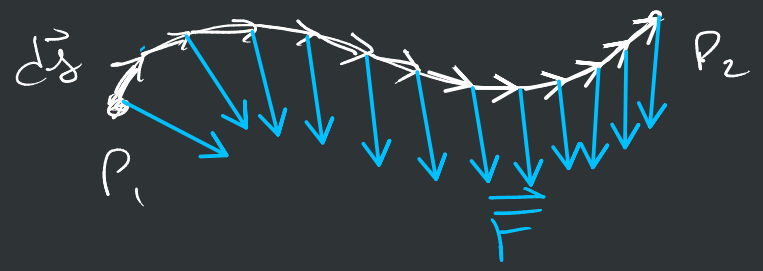
Chapter 2

$$W_{ME} = -\int_{P_1}^{P_2} \vec{F} \cdot d\vec{s}$$



$$\vec{F} = q\vec{E}$$

$$W_{ME} = -\int_{P_1}^{P_2} q\vec{E} \cdot d\vec{s}$$

$$\frac{W_{ME}}{q} = -\int_{P_1}^{P_2} \vec{E} \cdot d\vec{s}$$

$\Delta U = W_{ME}$

$\Delta U = -W_{\vec{E}}$

$\Delta K = 0$

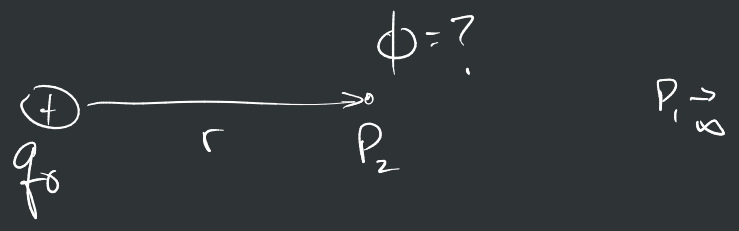
book notation

$$\frac{U(P_2) - U(P_1)}{q} = \frac{\Delta U}{q} = \Delta\phi = \phi_{21} = -\int_{P_1}^{P_1} \vec{E} \cdot d\vec{s}$$

electric potential difference (my notation)

$\downarrow$

$\phi(P_2) - \phi(P_1)$



$$\frac{U}{q} = \phi \qquad \left[\frac{\text{Joule}}{\text{Coulomb}}\right] = [\text{Volt}]$$

Along a closed path $W_{ME} = 0$

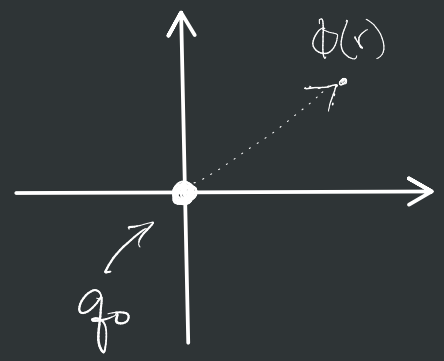
$\Delta U = 0$

$\Delta \phi = 0$

$$\oint \vec{E} \cdot d\vec{s} = 0$$

Ex: $\phi(r)$ for a point charge located at the origin.

$$\phi = -\int_{P_1 \leftarrow \infty}^{P_2 \leftarrow r} \vec{E} \cdot d\vec{s}$$



$$\phi = -\int_{\infty}^{r} \frac{kq_0}{r^2}\hat{r}\cdot dr\,\hat{r}$$

$$\phi = -kq_0 \underbrace{\int_{-\infty}^{r} r'^{-2}\,dr'}_{(-1)r'^{-1} = -\frac{1}{r'}\Big|_{\infty}^{r} = -\frac{1}{r} + \frac{1}{\infty}}$$

$$= -\frac{1}{r}$$

$$\phi(r) = +\frac{kq_0}{r}$$

$$\phi(r) = \frac{q_0}{4\pi\epsilon_0 r}$$

$$U(r) = \frac{q_0 q_1}{4\pi\epsilon_0 r} \qquad \phi = \frac{U}{q_1} = \frac{q_0}{4\pi\epsilon_0 r}$$

✓

<u>Ex:</u> Find ϕ inside + outside a uniformly charged sphere.

ρ

<u>outside</u> $r > R$

$$\phi = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r}$$

$$E = \frac{\rho R^3}{3\epsilon_0 r^2} = \frac{Q}{4\pi\epsilon_0 r^2} \qquad \underline{Q = \rho\,\tfrac{4}{3}\pi R^3}$$

$$\phi(r)_{\text{outside}} = \frac{Q}{4\pi\epsilon_0 r} = \frac{\rho R^3}{3\epsilon_0 r}$$

inside $r < R$

$$E = \frac{\rho r}{3\epsilon_0} \qquad \frac{3Qr}{4 \cdot 3\epsilon_0 \pi R^3} = \frac{Qr}{4\pi\epsilon_0 R^3}$$

$$\phi = -\int \frac{\rho r}{3\epsilon_0} \cdot dr$$

$$\phi(r) = \underbrace{-\int_{-\infty}^{R} E_{out}\, dr}_{} - \int_{R}^{r} E_{inside} \cdot dr$$

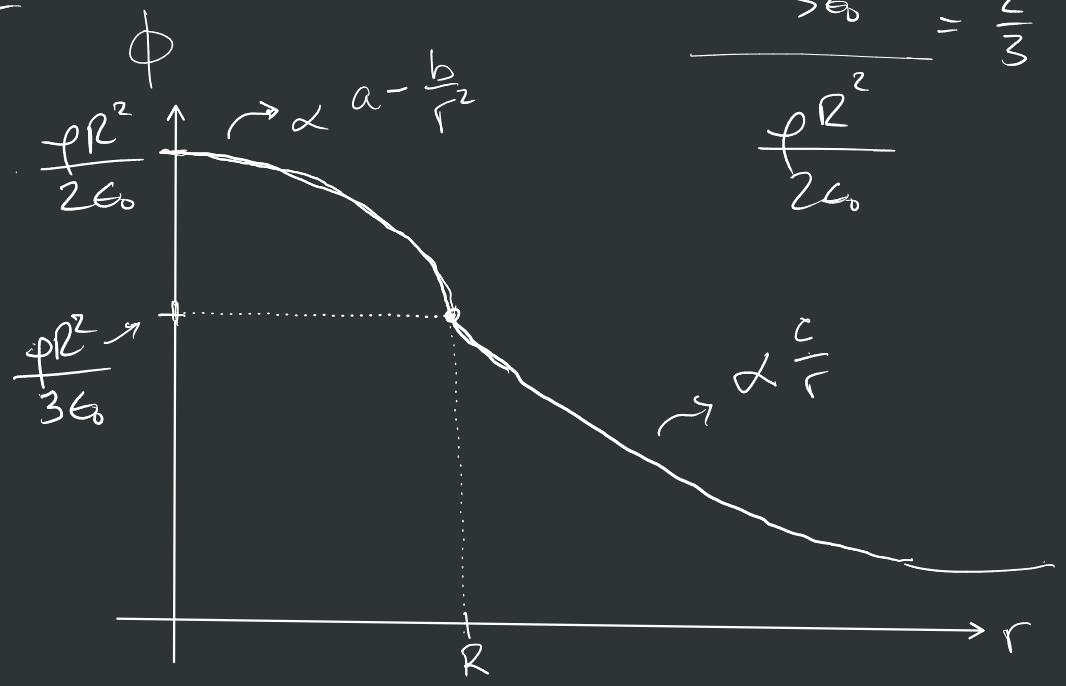
$$\phi(r) = \frac{\rho R^{\cancel{3}\,2}}{3\epsilon_0 \cancel{R}} - \int_{R}^{r} \frac{\rho r'}{3\epsilon_0} dr'$$

$$= \frac{\rho R^2}{3\epsilon_0} - \frac{\rho}{3\epsilon_0} \cdot \frac{1}{2} r^2 \Big|_R^r$$

$$= \frac{\rho R^2}{3\epsilon_0} - \frac{\rho}{6\epsilon_0}(r^2 - R^2)$$

$$\frac{2}{2}\frac{\rho R^2}{3\epsilon_0} + \frac{\rho R^2}{6\epsilon_0} - \frac{\rho r^2}{6\epsilon_0}$$

$$\phi(r)_{inside} = \frac{\rho R^2}{2\epsilon_0} - \frac{\rho r^2}{6\epsilon_0}$$



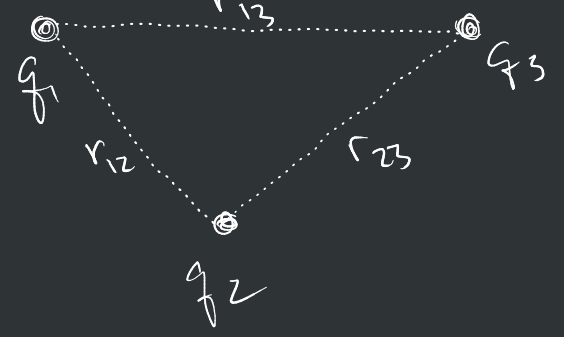
$$\frac{\frac{\rho R^2}{3\epsilon_0}}{\frac{\rho R^2}{2\epsilon_0}} = \frac{2}{3}$$

$$\left(\frac{1}{2} - \frac{1}{6}\right)\frac{\rho R^2}{\epsilon_0}$$

$$\frac{3}{6} - \frac{1}{6}$$

$$\frac{2}{6} = \frac{1}{3}$$

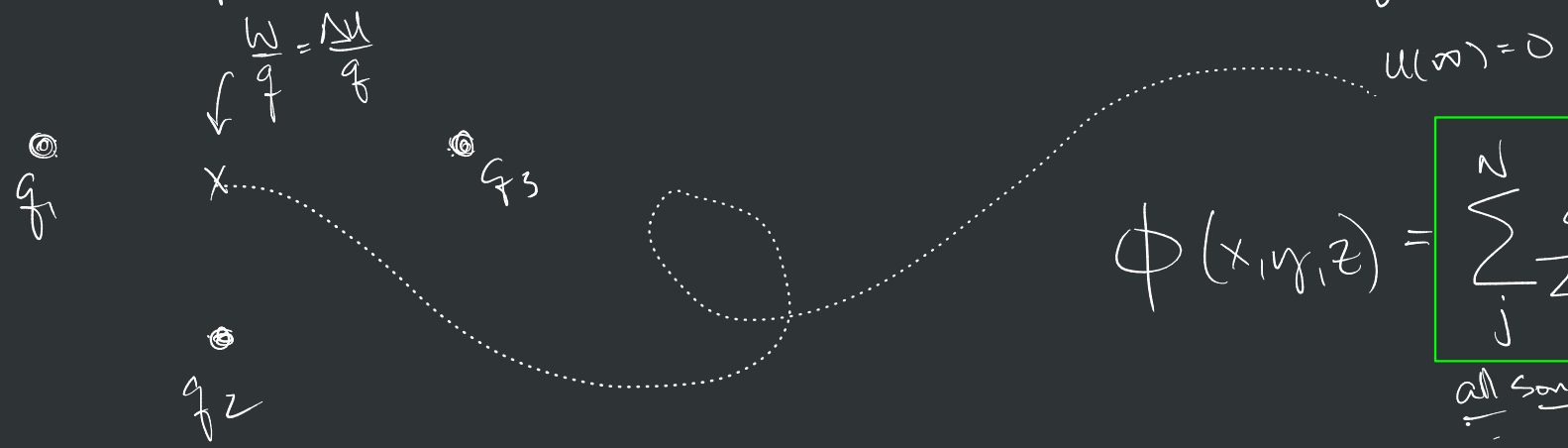
- Potential energy of a collection of point charges:



$$W_{ME} = U = \frac{1}{4\pi\epsilon_0}\left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}}\right)$$

$$U = \frac{1}{2}\sum_{j}^{N}\sum_{k\neq j}^{N}\frac{1}{4\pi\epsilon_0}\frac{q_j q_k}{r_{jk}}$$

- <u>Electric potential</u> around a collection of charges:



$$\phi(x,y,z) = \boxed{\sum_{j}^{N}\frac{q_j}{4\pi\epsilon_0 r_j}}$$

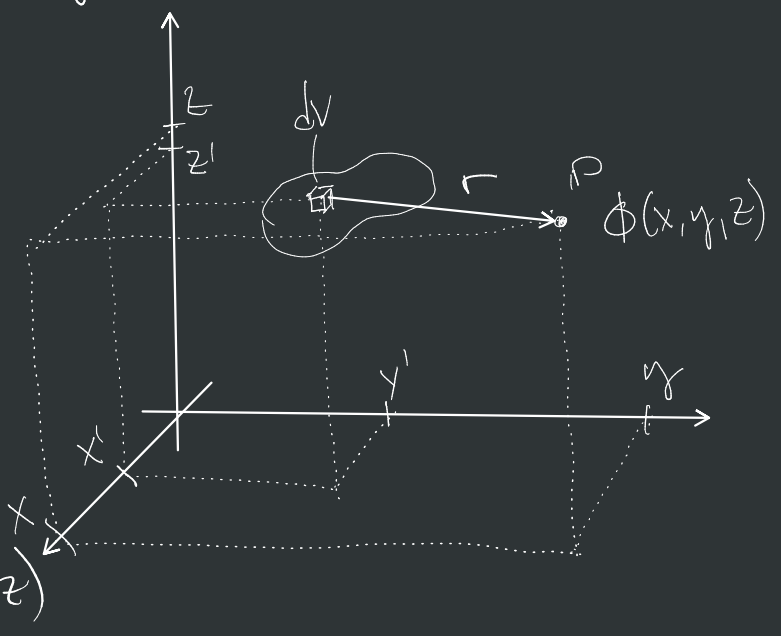
<u>all sources</u>

- Electric potential around a continuous charge distribution

$$q_j \to dq \to \rho\, dV$$

$$\phi(x,y,z) = \int \frac{\rho(x',y',z')\,dx'\,dy'\,dz'}{4\pi\epsilon_0 r}$$

r is distance from (x',y',z') to (x,y,z)



- Relationship between the work (potential energy) to build a continuous distribution and the electric potential of that distribution

$$U = \frac{1}{2}\sum_{j}^{N}\sum_{k\neq j}^{N}\frac{1}{4\pi\epsilon_0}\frac{q_j q_k}{r_{jk}}$$

$$U = \frac{1}{2}\underbrace{\boxed{\sum_{j}^{N} q_j}}_{\rho}\ \underbrace{\boxed{\sum_{k\neq j}\frac{1}{4\pi\epsilon_0}\frac{q_k}{r_{jk}}}}_{\phi}$$

continuous

$$\boxed{U = \frac{1}{2}\int \rho\,\phi\,dv}$$ eq. 2.32

source volume

another way

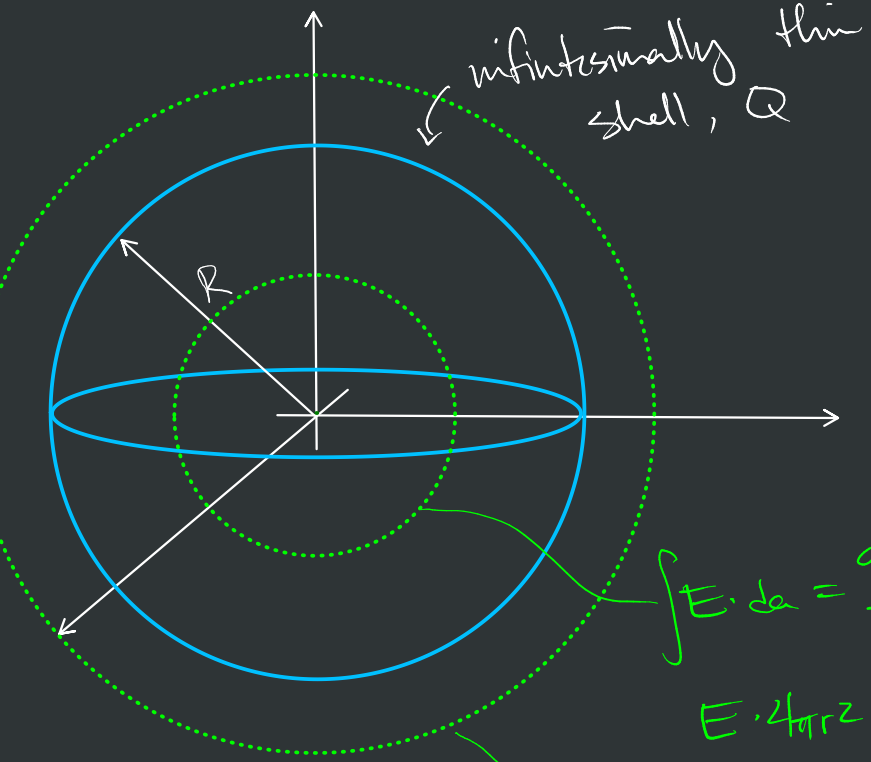
$$U = \frac{\epsilon_0}{2}\int_{\text{entire field}} \vec{E}\cdot\vec{E}\,dv$$

$$\vec{E}\cdot\vec{E} = E^2$$

$$\boxed{U = \frac{\epsilon_0}{2}\int_{\text{entire field}} E^2\,dv}$$

Chapter 1 eq. 1.53

Problem 1.32



$\vec{E} = ?$

$\phi = ?$

$U = ?$

$$\int E \cdot da = \frac{q_{encl}}{\epsilon_0} = 0$$

$$E \cdot 4\pi r^2 = 0$$

$$E = 0$$

inside

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2}\hat{r}$$

outside

$$U = \frac{\epsilon_0}{2}\int \rho\, \phi\, dV$$

$$\rightarrow U = \frac{\epsilon_0}{2}\int_{\text{entire field}} E^2\, dV$$

$$U = \underbrace{\frac{\epsilon_0}{2}\int_{\text{field inside}} E^2\, dV}_{0} + \underbrace{\frac{\epsilon_0}{2}\int_{\text{field outside}} E^2\, dV}$$

$$dV = r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$U = \frac{\epsilon_0}{2}\int_R^{\infty}\int_0^{\pi}\int_0^{2\pi}\left(\frac{Q}{4\pi\epsilon_0 r^2}\right)^2 r^2 \sin\theta\, dr\, d\theta\, d\phi$$

$$\boxed{\int_0^{2\pi} d\phi = 2\pi}$$

$$\boxed{\int_0^{\pi} \sin\theta\, d\theta = 2}$$

$$U = \frac{\cancel{\epsilon_0}}{2}\left(\frac{Q^2}{4^2\pi^2\epsilon_0^{\cancel{2}}}\right)(\cancel{4\pi})\int_R^\infty \frac{1}{r^2}\,dr$$

$$\int_0^\pi \sin\theta\, d\theta$$

$$-\cos\theta\Big|_0^\pi$$

$$-(-1)-(-(+1))$$

$$+1+1=2$$

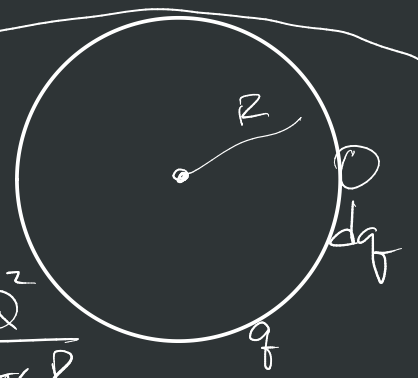
$$U = \frac{Q^2}{8\pi\epsilon_0}(-1)\left(\frac{1}{r}\right)_R^\infty$$

$$U = \frac{-Q^2}{8\pi\epsilon_0}\left[\cancel{\frac{1}{\infty}}^{\,0} - \frac{1}{R}\right]$$

$$\boxed{U = \frac{Q^2}{8\pi\epsilon_0 R}}$$

Find U by adding dq layer at R distance until you have Q.

$$U = \int_0^U dU$$



$$dU = \frac{k\,q\,dq}{R}$$

$$U = \int_0^Q \frac{q\,dq}{4\pi\epsilon_0 R} = \frac{q^2}{8\pi\epsilon_0 R}\Big|_0^Q = \frac{Q^2}{8\pi\epsilon_0 R}$$

Find ϕ.

$$\phi = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{r} \frac{Q}{4\pi\epsilon_0 r'^2}\,dr' = \frac{Q}{4\pi\epsilon_0 r'}\Big|_{\infty}^{r} = \frac{Q}{4\pi\epsilon_0 r}$$

$$\boxed{\phi(r) = \frac{Q}{4\pi\epsilon_0 r}} \qquad r > R$$

$$\phi(R) = \frac{Q}{4\pi\epsilon_0 R}$$

$$\phi = -\int_{\infty}^{r} \vec{E}\cdot d\vec{r} = -\int_{\infty}^{R} \vec{E}_{r>R}\cdot d\vec{r} - \underbrace{\int_{R}^{r} \overbrace{\vec{E}_{r<R}}^{=0}\, d\vec{r}}_{0}$$

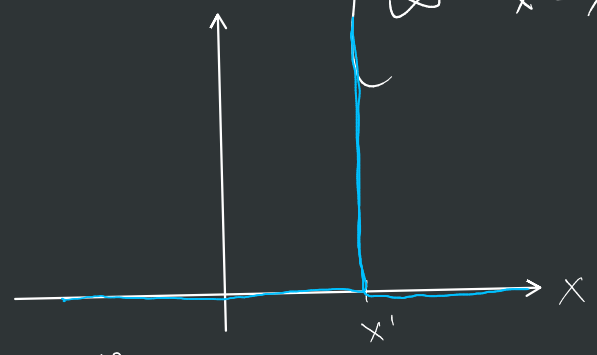
$$= \frac{Q}{4\pi\epsilon_0 R} - 0$$

$$\phi(r) = \frac{Q}{4\pi\epsilon_0 R} \qquad r < R$$

Find $U = \frac{1}{2}\int \rho\,\phi\,dv$.

$\rho = ?$ $\qquad \rho = \dfrac{Q}{V}$ $\quad Q \to \infty,\ V \to 0$ $\longrightarrow$ Dirac Delta

$$\delta(x'-x) = \begin{cases} 0 & x \neq x' \\ \infty & x = x' \end{cases}$$



$$\int_{-\infty}^{\infty} \delta(x'-x)\,dx = 1$$

$$\int_{-\infty}^{\infty} f(x)\,\delta(x'-x)\,dx = f(x')$$

$$Q = \int \rho(r)\,dV$$

$$\boxed{\rho(r) = \rho\,\delta(R-r)}$$

$$Q = \int_{\phi=0}^{2\pi}\int_{\theta=0}^{\pi}\int_{r=0}^{\infty} \rho\,\delta(R-r)\,r^2 \sin\theta\,dr\,d\theta\,d\phi$$

$$Q = 4\pi\rho \int_0^{\infty} r^2\,\delta(R-r)\,dr$$

$$Q = 4\pi R^2 \rho \longrightarrow \rho = \frac{Q}{4\pi R^2}$$

surface charge density

$$U = \frac{1}{2}\int \rho\,\delta(R-r)\,\Phi(r)\,dV$$

$$U = \frac{1}{2}\int_0^{2\pi}\int_0^{\pi}\int_0^{\infty} \rho\,\delta(R-r)\,\Phi(r)\,r^2 \sin\theta\,dr\,d\theta\,d\phi$$

$$U = \frac{1}{2}(4\pi)\int_0^{\infty} \rho\,r^2\,\Phi(r)\,\delta(R-r)\,dr$$

$$U = \frac{1}{2}\cdot(4\pi)\cdot\rho\cdot R^2\,\Phi(R)$$

$$U = \frac{1}{2}(4\pi)\cdot\rho\cdot R^2\cdot\frac{Q}{4\pi\epsilon_0 R}$$

$$\frac{Q}{4\pi R^2}$$

$$U = \frac{1}{2}\cancel{(4\pi)}\,\frac{Q}{\cancel{4\pi R^2}}\cdot\cancel{R^2}\cdot\frac{Q}{4\pi\epsilon_0 R}$$

$$\boxed{U = \frac{Q^2}{8\pi\epsilon_0 R}}$$

$$\phi = -\int_a^b \vec{E} \cdot d\vec{s} \qquad \cancel{\frac{d\phi}{ds} = -E}$$

Go to cartesian coordinates

$$\phi(x, y, z) \longrightarrow \frac{\partial \phi}{\partial x}, \quad \frac{\partial \phi}{\partial y}, \quad \frac{\partial \phi}{\partial z}$$

into a vector

$$\frac{\partial \phi}{\partial x}\hat{x} + \frac{\partial \phi}{\partial y}\hat{y} + \frac{\partial \phi}{\partial z}\hat{z} \longrightarrow$$ points in the direction of steepest change.

This vector is called the <u>gradient</u>

$$\vec{\nabla}\phi = \frac{\partial \phi}{\partial x}\hat{x} + \frac{\partial \phi}{\partial y}\hat{y} + \frac{\partial \phi}{\partial z}\hat{z}$$

$$\vec{E} = -\vec{\nabla}\phi$$